Google Cloud H2S

AI Builder Cup

Build in JAPAC.
Build for the world.



Team Details

a. **Team name: NexMind**
b. **Team leader name: Jeevan V**
c. **Problem Statement:**

Sometimes danger isn't loud yet — someone nearby just feels wrong: a stranger watching you on a train, a presence at the office after hours, a visitor whose behaviour turns, a partner in the room. You sense it before anything happens, but dialling 112 / 911, shouting or opening an SOS app would reveal you and provoke the very person you fear. QuickBite is a real food-delivery app that is secretly a two-way emergency line. A Gemini Live “restaurant employee” turns a whispered food order into a live, located incident for responders and guides you to safety, without ever breaking the disguise.

Theme: Sustainability & Social Impact · resilience strengthening · community support

Brief about the idea

The people who most need help often can't be seen asking for it

THE SCALE

1 in 3 women experience partner or sexual violence in their lifetime
WHO, 2026

10 min one woman or girl is killed by a partner or family member, every 10 minutes
UNODC / UN Women, 2025

4.41 lakh crimes against women recorded in India in 2024
NCRB, 2026

< 40% of women facing violence seek any help; under 10% go to police
UN DESA

ONE CALL, TWO CONVERSATIONS

1. One of many: Priya's partner is in the room, watching her phone. (Same flow fits a train, a late office, an unsafe visit.)
2. She opens QuickBite and taps “Call to order”.
3. Mia, the Gemini Live “restaurant”, answers and takes the order:

> ***"Would you like extra spicy, extra pepperoni, or garlic bread on the side? Extra spicy means someone is hurting you right now, extra pepperoni means a weapon is near you, garlic bread means you are being followed."***

He hears dinner being ordered. Responders see a located, high-severity incident fill in field by field, then Mia guides Priya to the nearest police station as “the rider’s pickup point”.

Disguised request any of 4 hidden ways in → **Gemini decodes live** codes → structured facts → **Responders act** map, route, video, evidence

Sources: WHO violence against women fact sheet (Jun 2026); UNODC/UN Women “Femicides in 2024” (Nov 2025); NCRB Crime in India 2024 (May 2026); UN DESA The World’s Women 2015. Priya is an illustrative caller.

Opportunities

Every other option has a tell. QuickBite looks like dinner.

How QuickBite differs from every existing option

● yes ◐ partly / needs setup or luck ○ no	Stays hidden from the abuser			Still gets real help		
	Screen looks innocent	Usable with abuser in the room	No code word to learn first	Responder can guide the caller	Responder sees location & details live	Live AI talks with the caller
Emergency SOS / 112 / 911	○	○	●	●	●	○
Himmat Plus	○	○	●	○	●	○
bSafe	◐	○	●	○	●	○
Aspire News	●	◐	●	○	○	○
Ask for Angela	◐	◐	○	●	○	○
RapidSOS HARMONY	○	○	●	●	●	●
Noonlight	○	○	●	○	●	○
ADT SoSecure	○	○	●	●	●	○
QuickBite	●	●	●	●	●	●

Why not just a code word? 911/112 bodies avoid published codes because abusers learn them too (NENA). Mia teaches the code inside the call, so it never has to be published.

Sources: NENA via CNN (Nov 2019); Aspire News, Himmat Plus, bSafe, Ask for Angela, RapidSOS HARMONY, Noonlight, ADT SoSecure public pages.

USP of the proposed solution

> **The only emergency line that looks like ordering dinner, and teaches its own code while you talk.**

No product we found combines **all six columns** (desk research, Oct 2026).

List of features offered by the solution

A complete loop: real situations in, one AI that listens, one console that acts

1 Four ways in

- Call to order: live call with Mia
- Click & order: coded cart
- Delivery instructions: silent taps + photo
- Heart double-tap SOS: silent AI observer

2 Mia, on Gemini Live

- Every code's meaning spoken with it
- 12 function-calling tools, live extraction
- 6 languages incl. Malayalam, Hindi, Tamil
- Voice stress, coercion, camera + sound

3 Getting to safety

- GPS with IP fallback; address geocoded
- Turn-by-turn route to police / hospital
- Directions disguised as “meet the rider”
- Open-mode first aid; messages via Mia

4 Responder console

- Ranked live queue, alerts, map case board
- Live transcript decoded, video, listen-in
- Case summary, linked cases, report email
- Analytics + AI Insights: hotspots, patterns, Gemini-written next actions

One AI, four ways it adapts

Mia reads the situation and switches mode — every path is live in the app.

● Covert call — you're watched

Abuse at home · Being hurt now · Weapon nearby · Taken / locked in · Witnessing a crime · Drug dealing · Illegal mining

Codes hidden inside a normal food order.

● Open mode — you can speak

Can talk freely · Fire / accident · Medical emergency

Mia drops the cover: plain questions + first aid.

● On the move — chased

Being followed · Being chased

Live turn-by-turn route to the nearest police station.

● Silent SOS — can't speak

Hostage · Abduction

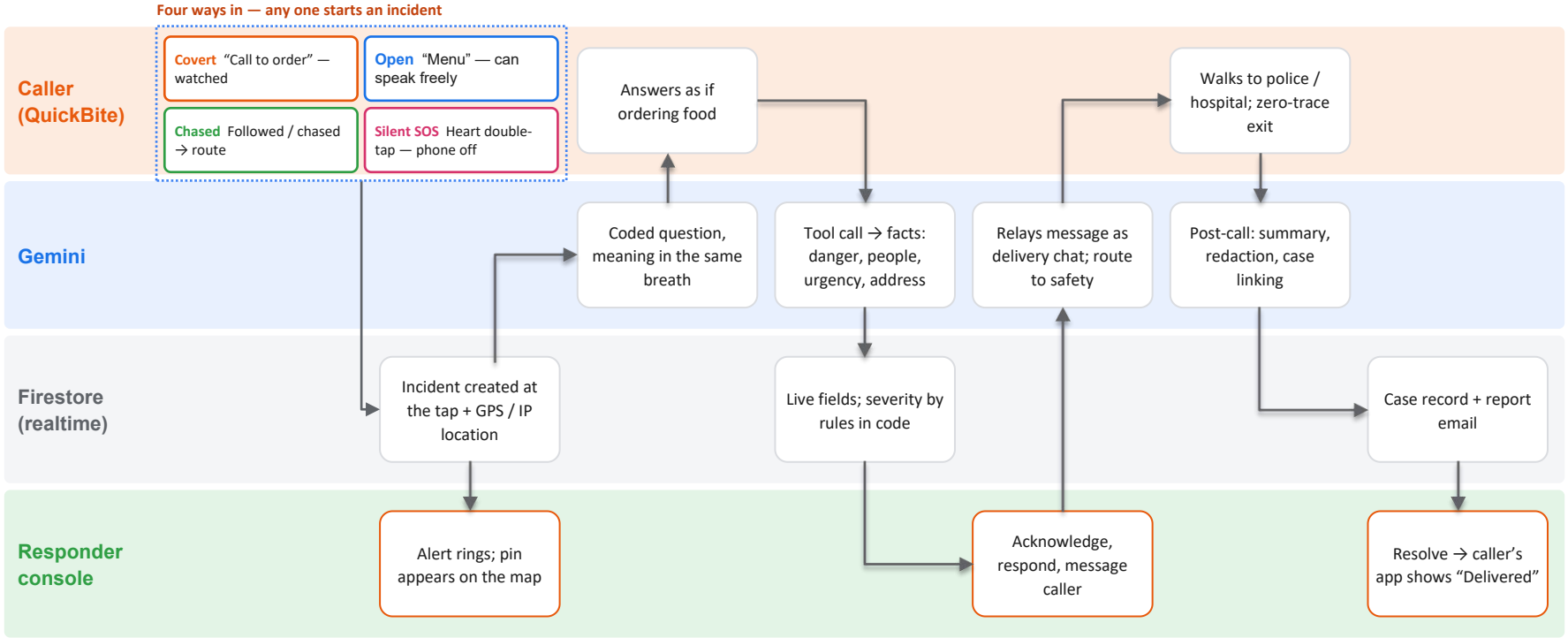
Heart double-tap: dark screen, silent AI observer.

Disguise & security: zero-trace exit · food-app wording on every prompt · strict Firestore rules, role-based access · practice call never reaches responders

All features above are built and running in the deployed web app.

Process flow diagram or Use-case diagram

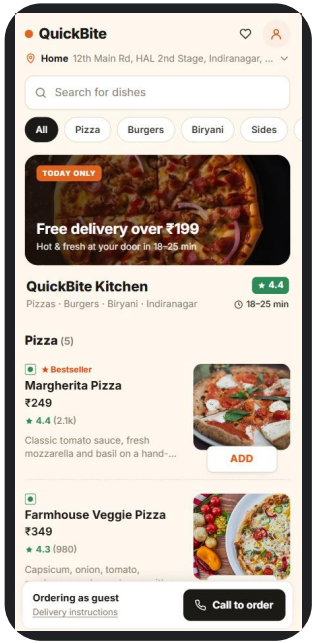
The incident exists before Priya finishes her first sentence



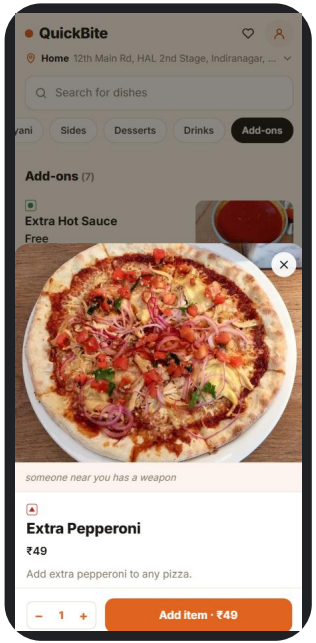
Use cases: caller in danger · witness ("for someone else" codes: crime seen, drug dealing, illegal mining, child in danger, fire) · responder · admin. **Measured:** danger tag on the dashboard before the caller's line is transcribed

Wireframes/Mock diagrams of the proposed solution (optional)

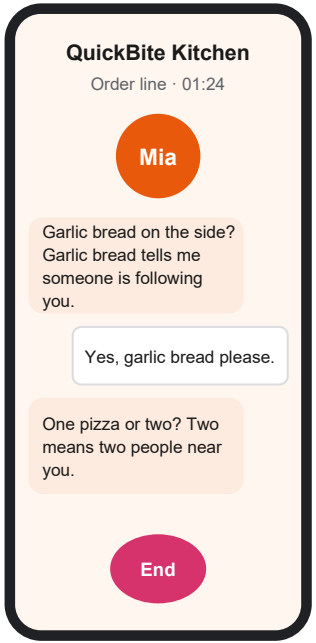
Every screen passes as a food app; every way to ask for help is an ordinary feature



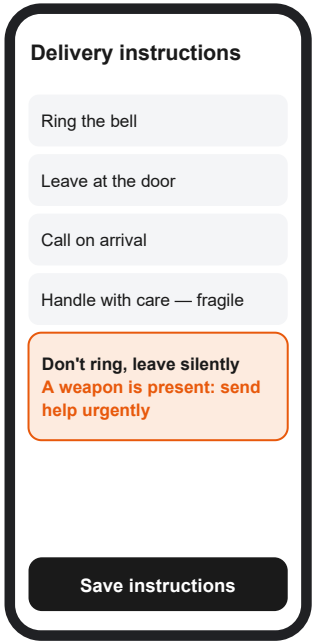
Home
Looks like any delivery app. The heart hides the SOS.



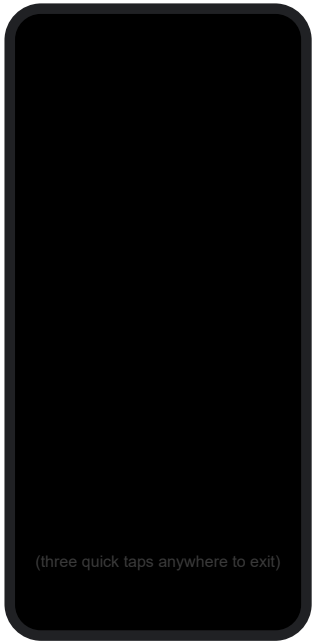
Item sheet
Long-press a coded item to see what it really means.



Call to order (mock)
Mia speaks every code's meaning with it.



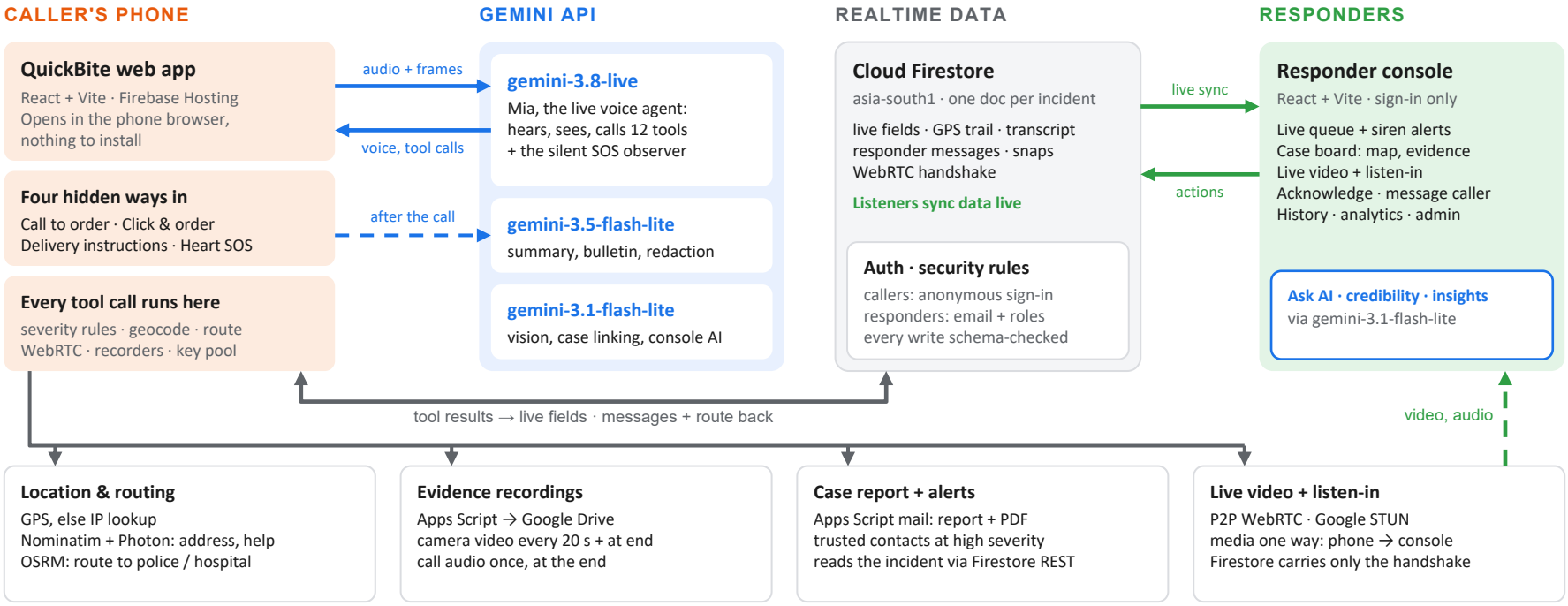
Delivery instructions (mock)
No sound needed: long-press reveals meaning.



Silent SOS (mock)
Screen looks off; cameras + AI observer run.

Architecture diagram of the proposed solution

Firebase backend: the phone, Gemini and Firestore do all the work



Managed Firebase backend · Gemini API keys: free tier first, paid fallback · AI usage metered per call · gitleaks secret scan · 4 Firebase Hosting sites

Full architecture docs on GitHub →

Technologies used in the solution

Built on Google: Gemini for intelligence, Firebase for realtime, the web for reach

Google AI	**Gemini Live API (gemini-3.8-live)** the only model doing realtime voice + vision + tools	**Gemini key pool** free key first, paid fallback, per model
	Gemini 3.5 / 3.1 Flash-Lite summaries, vision, linking, search, insights	**Gemma 4 31B** free judge in the eval harness
Firebase & Google Cloud	**Hosting (4 sites), Cloud Firestore, Auth** email, device ID, phone OTP	**Apps Script + Drive** evidence and mail, no billing
	Firestore security rules role-based responder access	**Maps JS, STUN** maps, WebRTC
Apps	**React + Vite + TypeScript** caller app, console, landing	**Web Audio + getUserMedia** mic, camera frames, audio mixing
	WebRTC (browser) live video + listen-in, P2P	**ECharts, Leaflet** analytics
Open map data	**OpenStreetMap, Nominatim, Photon** geocoding, landmarks, nearby help	**OSRM** turn-by-turn routing
Quality & DevOps	**GitHub Actions** gitleaks secret scan	**Staging site + preview channels** safe testing before production
	Performance test (Node + Python) 8 real calls measured from incident records, Excel report	

Estimated implementation cost (optional)

Built on free tiers for $0, about $120–180 a month to run a district pilot

1 · PROTOTYPE (DONE)

$0

infrastructure spend to build

- **Team:** 2 developers, ~4 weeks
- **Hosting, Firestore, Auth:** Firebase free tier
- **AI:** Gemini free key; $0 billed across 59 real test calls
- **Maps, routing, mail:** OpenStreetMap, OSRM, Apps Script, all free

2 · PRODUCTION SCALE-UP (EST.)

≈ $60–120

per month, fixed

- **Firebase Blaze:** paid Firestore + Hosting
- **Cloud Run + ADK:** keys server-side, dispatch agents
- **Google Maps Platform:** geocoding, places, routes
- **TURN relay:** video on strict networks

3 · RUNNING COST (EST.)

≈ $0.20

of AI per call, paid prices

- **Measured:** app usage meter, 59 real calls; 98% is the live call
- **District pilot (rough estimate):** assuming ~300 calls a month (10/day) × $0.20 ≈ $60 a month
- **No new staff:** incidents go to existing 112 / helpline responders

COST LEVERS ALREADY BUILT

~22 → 5
model inferences per call, now 5–6 for 4 turns (silent tool replies)

0.5 fps
low-res camera frames (~70 tokens each)

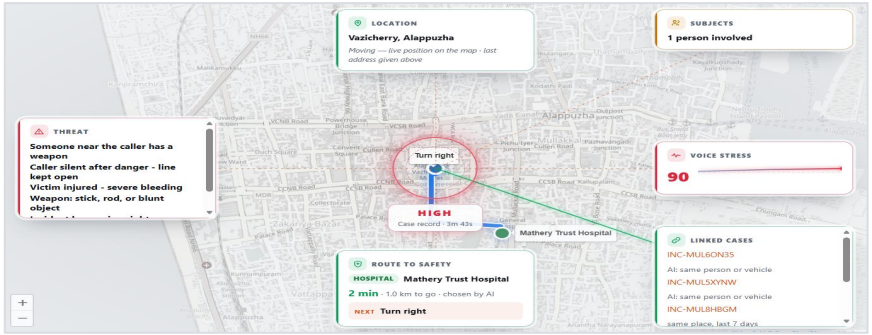
Per task
smallest model; free tier first

52 → 25 KB
Mia's prompt rewritten: about half the tokens per reply

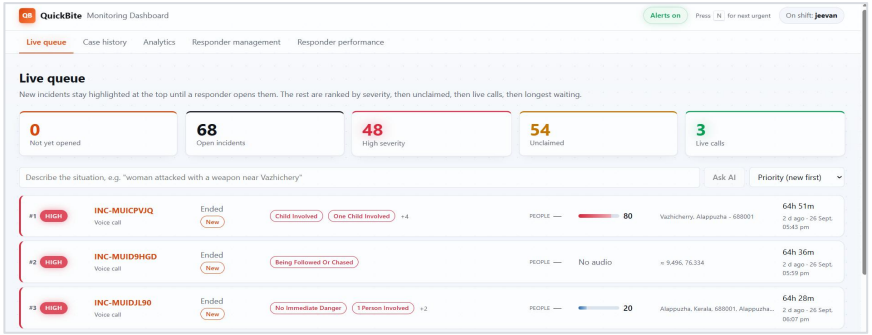
AI cost: Data + calculations: [QuickBite AI cost workbook (Google Sheets)]

Snapshots of the prototype

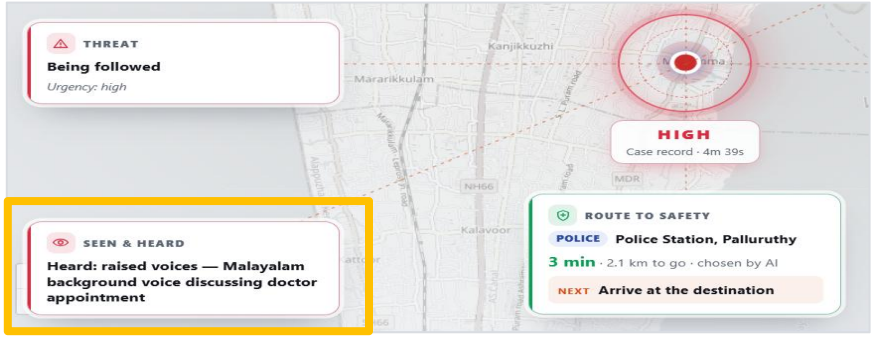
Live and deployed: what a responder sees while Priya is still on the call



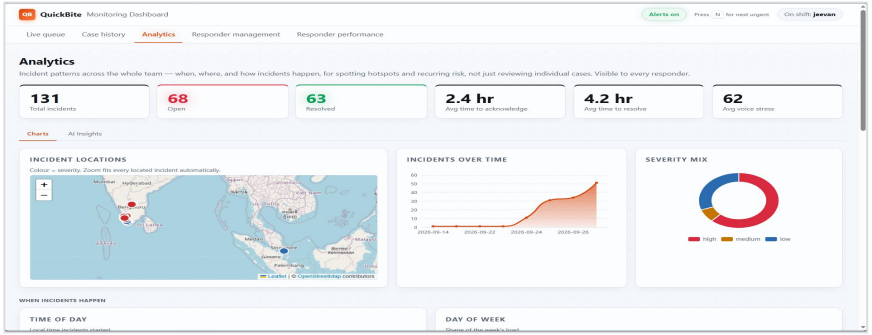
Case board: threat, voice stress, linked cases, live route — all filled by Gemini



Live queue: severity-ranked incidents, alerts, AI search



Seen & heard: background voices logged as evidence



Analytics + AI Insights: hotspots, patterns & Gemini-written next actions across all incidents

Prototype Performance report/Benchmarking

8 real voice calls on the deployed app: danger is flagged while the caller is still speaking

2–3.5 s early
danger and coercion tags reach the dashboard before the caller's line is transcribed

~535 m
route to the police station tracked live in every chase run

$0.20
AI cost per call (median, paid-tier prices)

16.8 s
from "being followed" to Mia's first turn-by-turn direction (median)

Danger flagged before the words appear

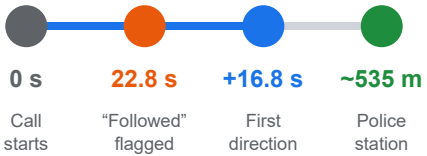
Seconds the danger tag beat the transcript, covert calls



Coercion (husband's voice) flagged the same way in all 4 calls.

Chase: followed → guided to safety

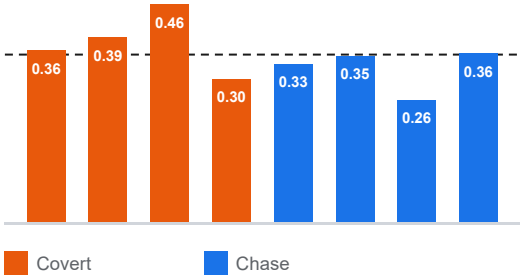
Median of 4 chase calls, from the incident timeline



Every run: route drawn live, turn-by-turn directions, arrival on the map.

AI cost per call

Paid-tier prices; median $0.35



Data + calculations: [QuickBite performance test (Google Sheets)]

Additional Details/Future Development (if any)

From prototype to pilot to platform

Next 3 months
Pilot-ready

- Even faster voice: pre-connect Gemini, per-turn latency metering
- Cloud Run + ADK agents: dispatch, verification; keys server-side
- Google Maps Platform for geocoding, places and routing, worldwide
- TURN relay for video on strict networks
- Full Android parity, then iOS

6–12 months
Real dispatch

- Alerting nearby app user for quick response.
- 112 / ERSS integration: hand decoded incidents to the state response centre
- Pilot with an NGO and police (e.g. Kerala Police Pol-App context)
- More languages for new regions
- Responder mobile app for field units

Platform
Beyond calls

- Bluetooth mesh relay for no-signal areas
- On-device Gemini Nano extraction for privacy and offline
- Anonymous whistleblowing channel on the same AI backbone
- Integration path into Android (next slide)

What we need next

A public-safety pilot partner (NGO or police unit) · Gemini API credits for a district pilot · Android Personal Safety team feedback on a Discreet mode · mentorship to take it to 112 integration

Kerala Police Pol-App and Nirbhaya Nisha are existing state programmes we would pilot alongside; no partnership exists yet.

Provide links to your:

Submission links

GitHub repository	github.com/Jeevanvsan/ai_builder_hack
Demo video	QuickBite demo: Covert Call (Google Drive) **Alternative:** stream in the browser on our landing page
Product link	quickbite-5cde0-landing.web.app

Extras

Chasing-scenario video	Someone chasing, guided to safety (Drive) **Alternative:** stream in the browser on our landing page
Cost workbook	AI cost: 59 real calls + estimate (Sheets)
Performance test	8 real calls: timings, route, cost (Sheets)
User manual	quickbite-5cde0-landing.web.app/manual

Recorded demo incidents

Real incidents the dashboard recorded in the demo videos, to verify (sign in first).

Covert call	INC-MUZNKM7N (covert order call)
Chase	INC-MUZKKVL1 (chased, guided to safety)

FOR JUDGES: TRY IT YOURSELF

Caller app: quickbite-5cde0.web.app

Responder dashboard: quickbite-5cde0-dashboard.web.app

Login: judge@quickbite.com / judge@Quickbite123

User manual: step-by-step guide to every feature

Mandatory: allow the microphone at both levels: the browser's site prompt for QuickBite, and the browser app's mic permission in your phone/OS settings. Without it the call can't start and the app returns to the menu.

Steps

1. Open the dashboard on a laptop and sign in with the login above.
2. Open the caller app on a phone or another tab; allow mic (required) and location.
3. Tap "Call to order". Mia asks if you can talk: say "order" (someone listening) or "talk".
4. Answer her food questions. Each code's meaning is spoken with it, e.g. "garlic bread" = someone is following you.
5. Watch the incident appear on the dashboard and fill in live: location, danger tags, transcript.
6. Hang up and open the incident: case summary, map, route and evidence.

Team: **NexMind** Members: **Al Ameen N, Jeevan V**

How Gemini powers QuickBite — and the path to a Google product

A real-time multimodal agent today; a Discreet mode inside Android tomorrow

NOT A CHATBOT — A VERIFIED MULTIMODAL AGENT

- **Real-time voice + vision:** Gemini Live listens and watches, and decides on its own which of 12 function-call tools to fire mid-sentence — an agent that acts, not a chatbot that replies
- **Live structured incident data:** a disguised conversation becomes danger, people and location, streamed to responders field by field
- **AI proposes, code decides:** deterministic rules set severity and launch the route to safety — no hallucinated life-or-death calls
- **Grounded & self-checking:** post-call summary, third-party redaction, case-linking, and a Gemma judge that validates every run

Danger reaches the responder's screen before the caller finishes speaking.

THE PATH TO A GOOGLE PRODUCT

- **Closes a real Android gap:** it can call for help silently, but not covertly with an attacker in the room
- **Ships as Discreet mode** in Android's Personal Safety — a Gemini Live covert persona inside Emergency SOS
- **Plugs into 112:** decoded incidents via ELS + Emergency Live Video; on-device Gemini Nano keeps extraction private
- **Firebase + Google Cloud** becomes the public-safety backend for agencies

Already proven on Google's stack: RapidSOS HARMONY runs Gemini Live + ADK in 911 centres.

Google Cloud H2S

AI Builder Cup

Build in JAPAC.
Build for the world.

Thank you

